Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Total Purchases

18

Data Columns

Comprehensive transaction data

37

Missing Values

Customer demographics & behavior

Only in Review Rating column

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Data Preparation Process

02

03

Data Loading
Initial Exploration
Missing Data Handling

Imported dataset using pandas

Structure check and summary statistics

Imputed Review Rating with median values

05

Feature Engineering
Database Integration

Created age groups and purchase frequency

Connected to PostgreSQL for analysis

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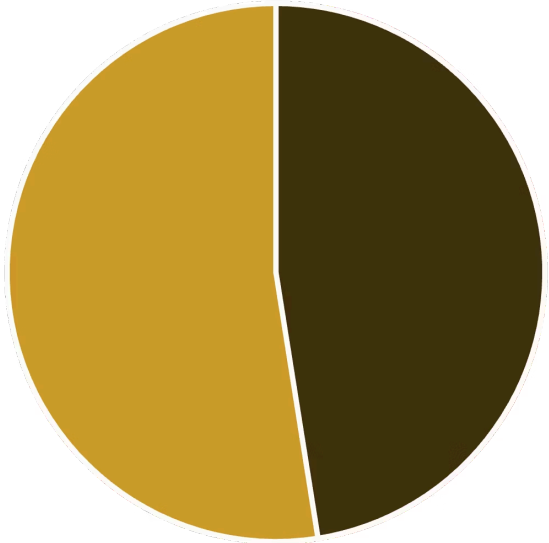
Revenue by Gender Analysis



Key Insight

Female customers generate slightly higher total revenue than male customers

Gender-based marketing strategies could optimize revenue streams

Male

Female

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High-Value Discount Users

Customers using discounts while spending above average

Smart Shoppers

High spenders who maximize value with discounts

Strategic Opportunity

Target premium customers with exclusive offers

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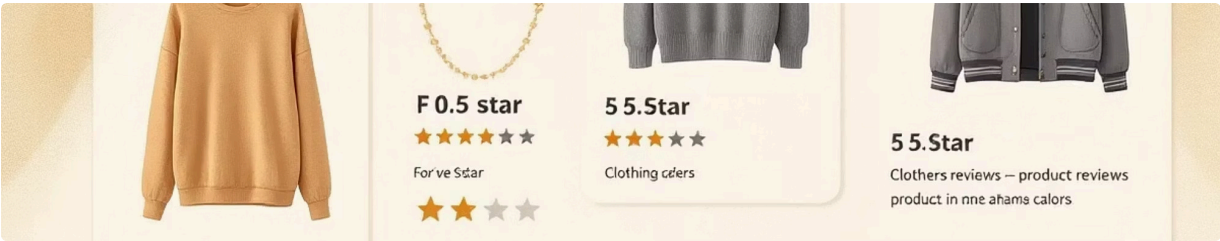

Top-Rated Products

5

5

Blouse
4
Dress
Shirt

Highest customer satisfaction
Consistently excellent reviews
Strong customer approval

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Shipping Preferences Impact

Express Shipping

Standard Shipping

$65

$58

Average purchase amount

Average purchase amount

Express shipping customers spend 12% more per transaction

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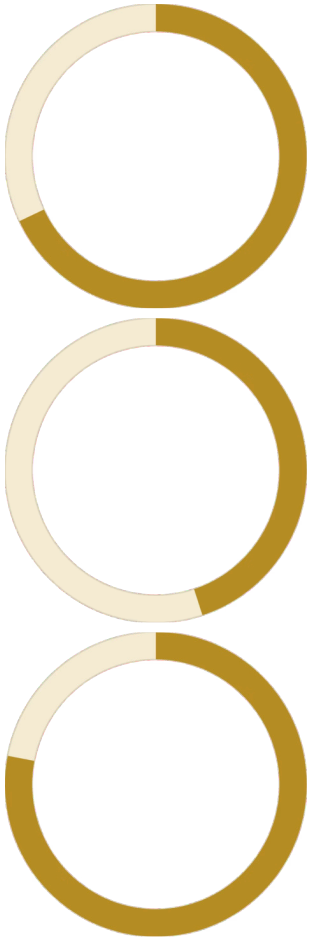
Subscription Impact



Higher Spend
Revenue Share

Subscribers vs non-subscribers

Loyalty Rate

From subscription customers

Repeat purchase frequency

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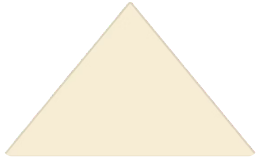
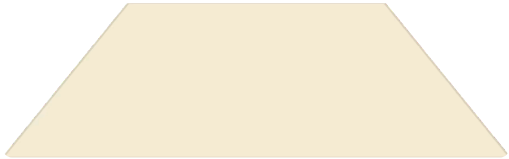
Customer Segmentation



Loyal

1

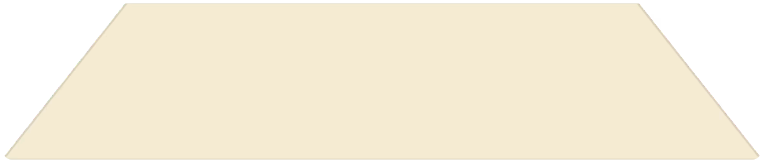
15% - High value customers



Returning

2

35% - Regular shoppers



New

3

50% - First-time buyers

Focus on converting New to Returning, Returning to Loyal

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Loyalty Programs

Reward repeat buyers to increase retention

Targeted Marketing

Focus on high-revenue segments and express users

Product Positioning

Highlight top-rated products in campaigns

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